1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, TLS/stromal engagement is common in MIBC but does not uniformly translate into better outcomes — why do some immune-infiltrated tumors still have poor survival? 2) Tier 1 paths used: I built on (a) the immunotherapy/immune-contexture insight (cross■cancer biomarker rationale), (b) MSH6 as a DNA■repair/prognostic anchor, and (c) the SARIFA tumor–adipocyte/stromal program that can suppress immune function. These three together define a puzzle of immune potential (TLS) versus molecular (MMR) and stromal brakes. 3) Tier 2 support: MDK/EV biomarkers, methylation markers (SDC2/TFPI2), metabolic/urinary signals (GLU, creatinine, microalbuminuria), and pZVR provide orthogonal evidence that (i) tumor cell programs and extracellular signals are measurable noninvasively and (ii) metabolic/stromal programs can correlate with aggressive, immune■excluded phenotypes. These strengthen the plausibility that a stromal program can modulate an otherwise immune■active tumor. 4) Why OS as endpoint: Overall survival is the most robust, clinically meaningful endpoint available across cohorts and best captures the net effect of interacting tumor, immune and stromal programs in MIBC. 5) Deliberate exclusions: I did not use the cross■cancer 'HPV status' as a primary anchor (it informed rationale only) because it is a generalizer rather than an MIBC■specific mechanistic node; I also deferred numerous Tier■3/metabolic ideas (HDL■C, exploratory glycan markers) to secondary_modifiers to keep the hypothesis tightly focused on one imaging feature plus two core anchors?

Working Hypothesis

TLS-rich tumors in MIBC gain a survival advantage only when MSH6 function is intact, enabling TLS-mediated antigen presentation to stimulate anti-tumor immunity. This advantage is abrogated by SARIFA-high stromal signaling that suppresses TLS activity and antigen presentation, so TLS-rich tumors lose survival benefit in the presence of SARIFA-high or MSH6 deficiency.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, SARIFA
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6 as a DNA■repair/prognostic anchor, and (c) the SARIFA tumor–adipocyte/stromal program that can suppress immune function.

SARIFA: SARIFA tumor–adipocyte/stromal program that can suppress immune function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MDK	Supporting modifier from knowledge graph

Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	GLU	Supporting modifier from knowledge graph
Secondary	EV-mRNA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and extracellular vesicle (EV) signals provide orthogonal readouts of the stromal and tumor-cell programs that may modulate TLS efficacy. OS is the most clinically meaningful endpoint to capture the net effect of tumor intrinsic factors, immune contexture, and stromal brakes on patient survival.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 3: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MDK: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SDC2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.